北京交通大學

信号与系统

School of Electronic
and Information Engineering

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2025.3

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System Frequency Domain Analysis

- Why Perform Frequency Domain Analysis of a System
- Frequency Domain Analysis of Continuous Time LTI Systems
- Frequency Domain Analysis of Discrete Time LTI Systems
- Amplitude Modulation and Demodulation of Continuous Signals
- Frequency Domain Analysis of System Using MATLAB

Amplitude Modulation and Demodulation of Continuous Signals

- ※ Why Modulate Signals
- ※ Basic Concepts of Signal Modulation
- ※ Amplitude Modulation of Continuous Signals
- ※ Synchronous demodulation of modulated signals
- ※ Single Sideband Amplitude Modulation Technique
- ※ Frequency Division Multiplexing for Signal Transmission

Why Modulate Signals

- In radio broadcasting systems, the size of the signal transmitting antenna is about one-tenth of the signal wavelength.
- The highest frequency of the music signal is 20kHz, which corresponds to a radio wavelength of 15 kilometers.
- Therefore, if the music signal is directly transmitted in the form of radio waves, it is difficult to realize.
- Move the low-frequency signal to a higher frequency band to facilitate wireless transmission.
- Different signals can use different frequency bands to make full use of the channel.

Basic Concepts of Signal Modulation

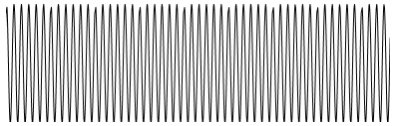
$x(t)$
Modulation Signal

Modulator

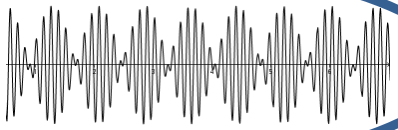
$y(t)$
Modulated Signal

Carrier Signal

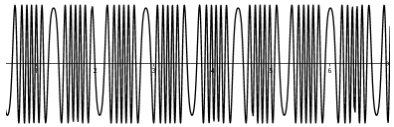
$$c(t) = A\cos(\omega t + \theta)$$



amplitude modulation (AM)

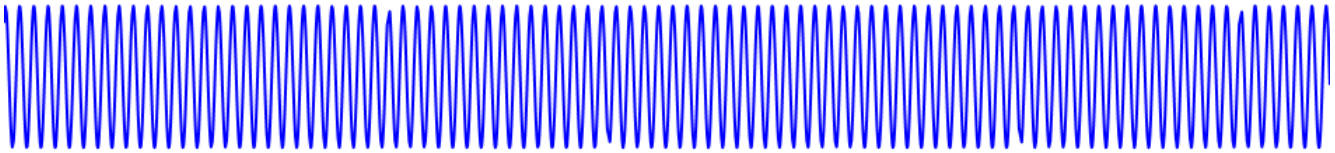


frequency modulation (FM)



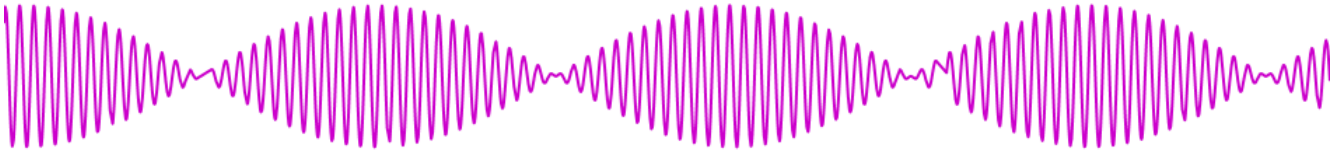
Basic Concepts of Signal Modulation

carrier signal

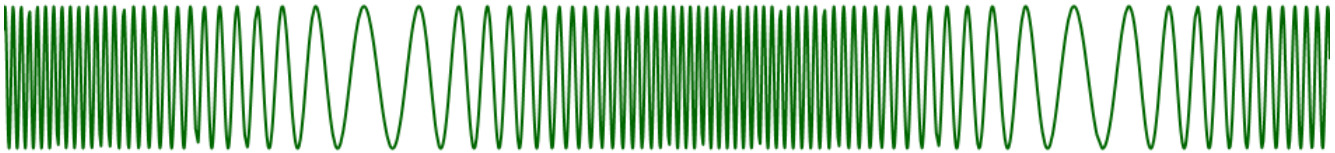


Modulated signal

amplitude modulation AM

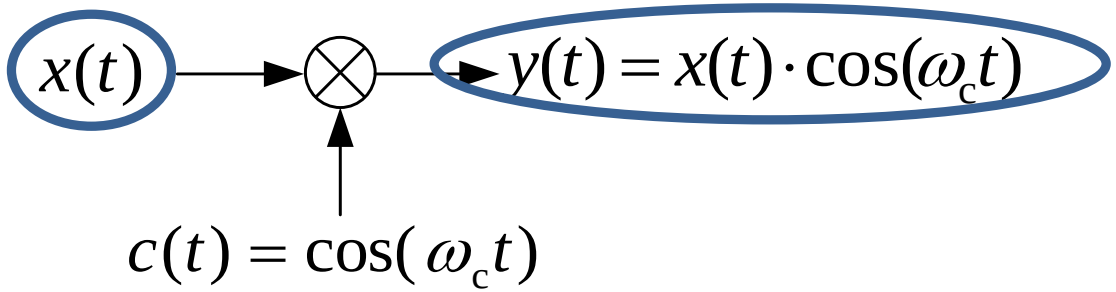


frequency modulation FM

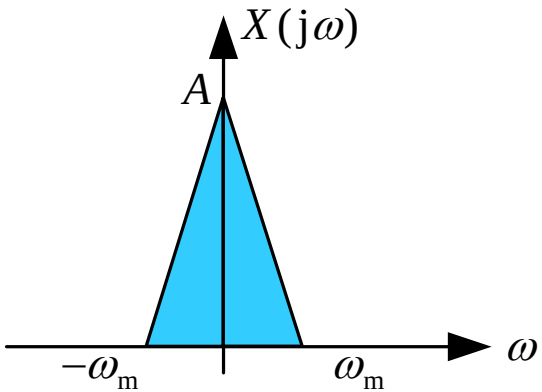


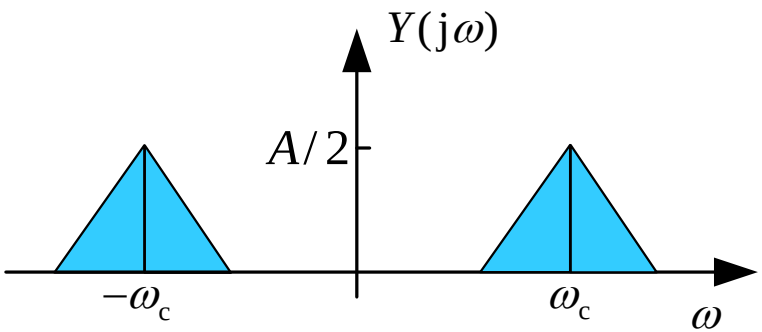
Amplitude Modulation of Continuous Signals

Amplitude Modulation Block Diagram



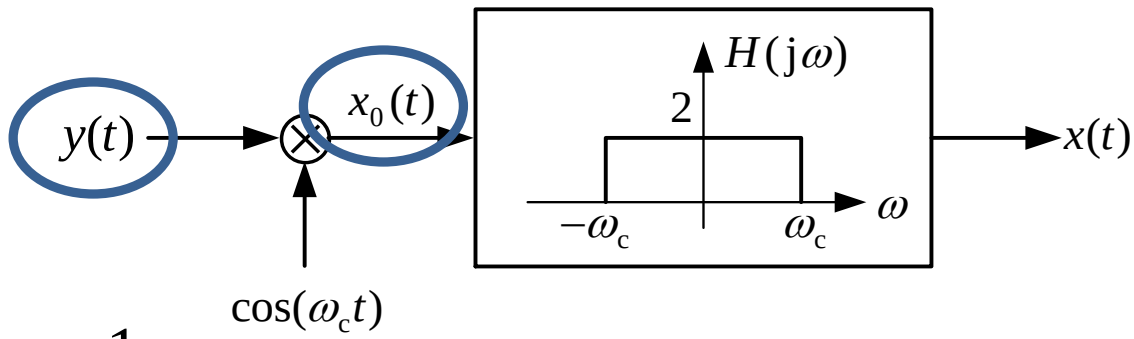
$$Y(\mathrm{j}\omega) = \frac{1}{2}\{X[\mathrm{j}(\omega+\omega_c)] + X[\mathrm{j}(\omega-\omega_c)]\}$$



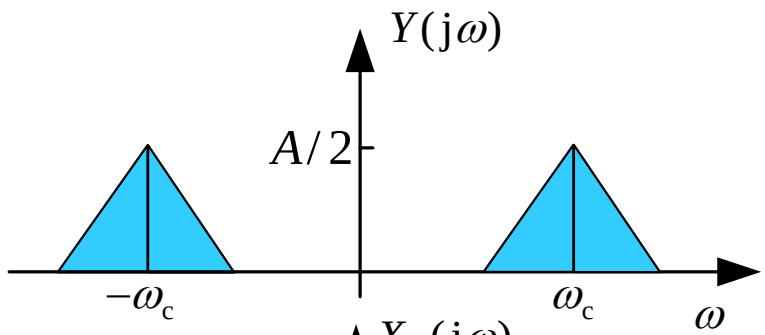


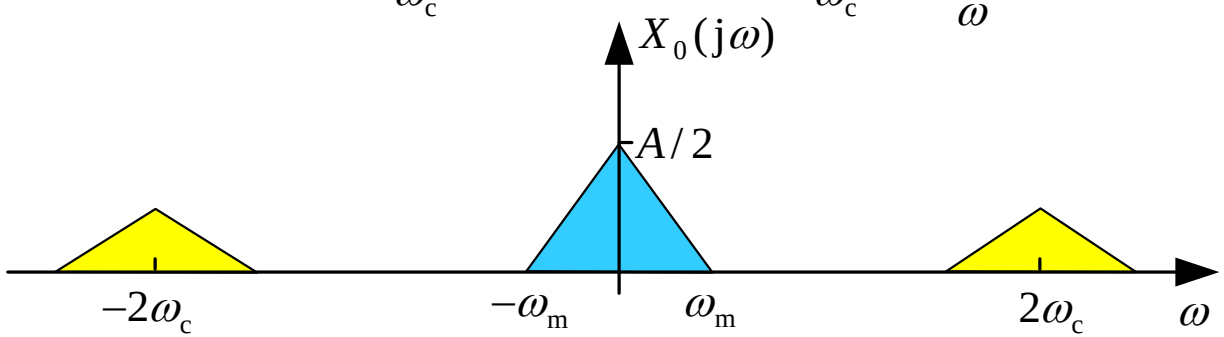
Synchronous demodulation of modulated signals

Block Diagram of Synchronous Demodulation



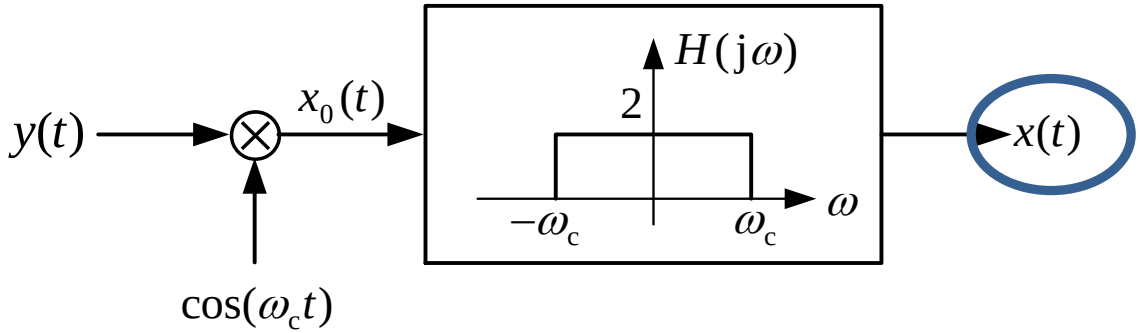
$$X_0(\mathrm{j}\omega)=\frac{1}{2}\{Y[\mathrm{j}(\omega+\omega_c)]+Y[\mathrm{j}(\omega-\omega_c)]\}$$





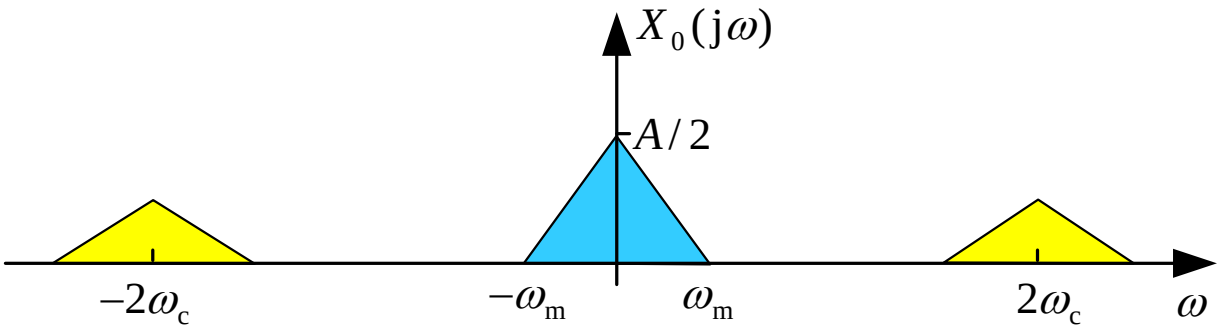
Synchronous demodulation of modulated signals

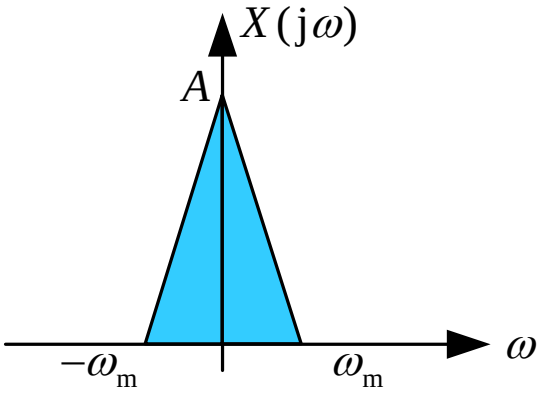
Block Diagram of Synchronous Demodulation



$$X_0(\mathrm{j}\omega)=\frac{1}{2}\{Y[\mathrm{j}(\omega+\omega_c)]+Y[\mathrm{j}(\omega-\omega_c)]\}$$

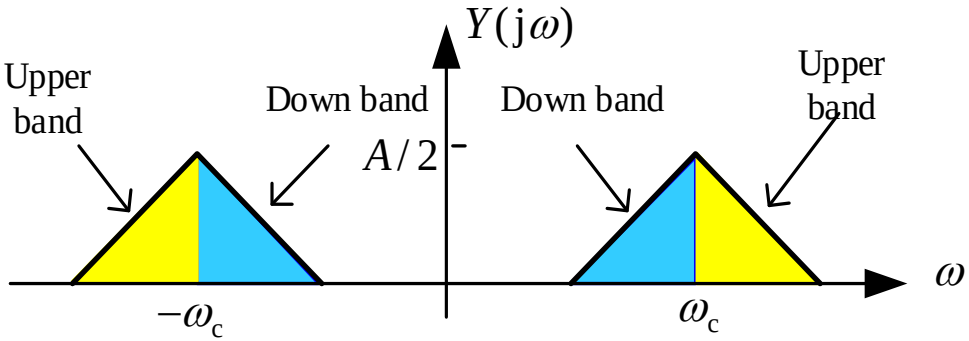
$$X(\mathrm{j}\omega)=X_0(\mathrm{j}\omega)H(\mathrm{j}\omega)$$



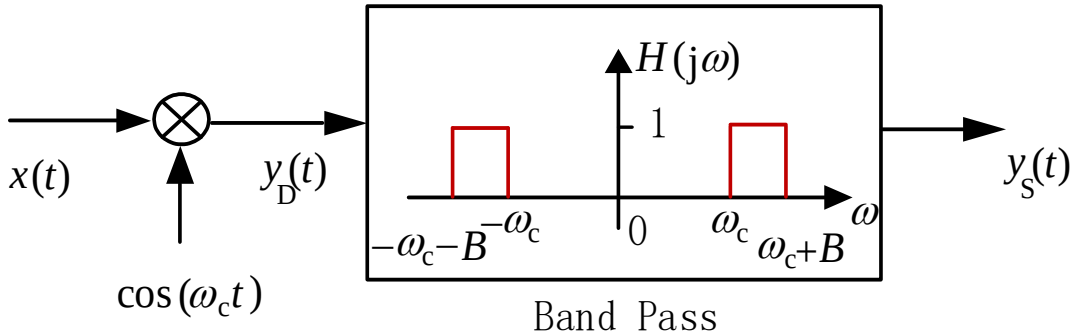


Unilateral band amplitude modulation technology

In order to improve the utilization rate of channel frequency bands, a unilateral zone amplitude modulation technology can be used



The principle box diagram with amplitude modulation above

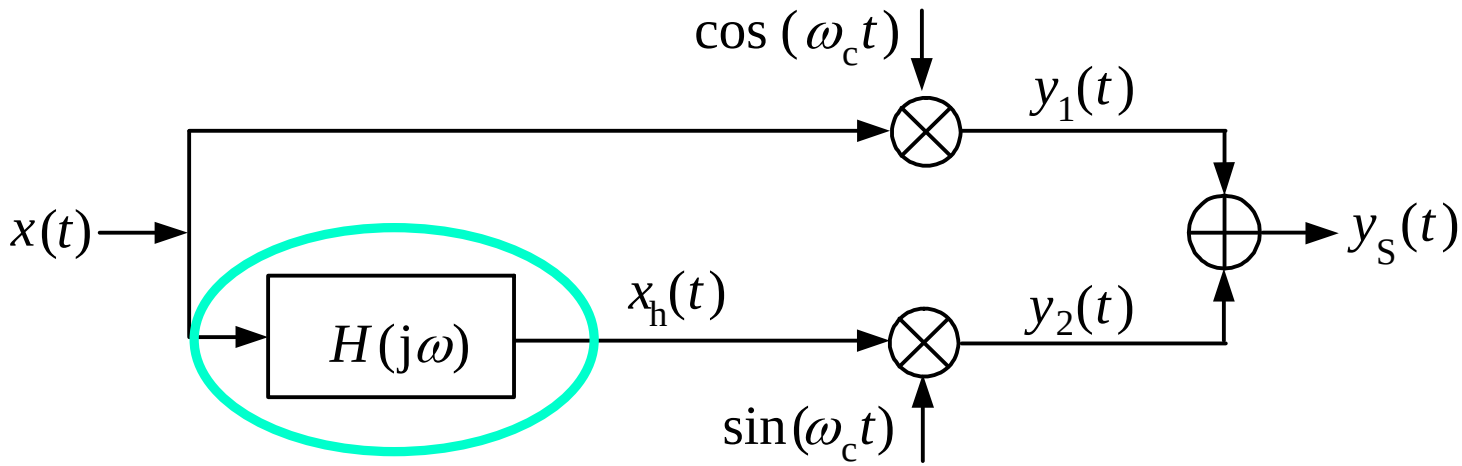


The frequency band of the rumor of the tape is appropriately changed

However, the ideal band filter cannot be achieved in engineering

Unilateral band amplitude modulation technology

In practical applications, the Hillbert transform device is often used to implement the signal unilateral zone modulation



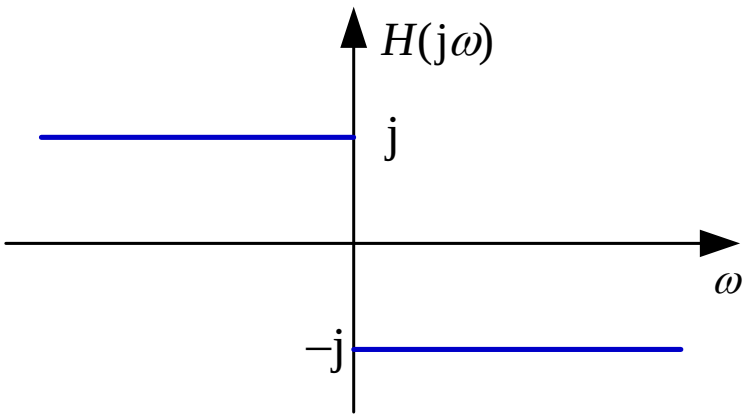
$$H(j\omega) = -j \cdot \mathrm{sgn}(\omega) = \begin{cases} -j, & \omega > 0 \\ j, & \omega < 0 \end{cases}$$

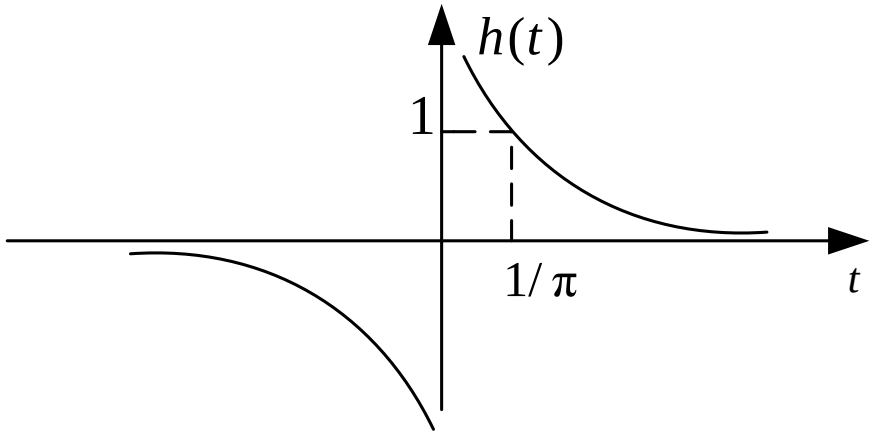
Unilateral band amplitude modulation technology

Hilbert converter

$$H(\mathrm{j}\omega) = -\mathrm{j}\cdot\mathrm{sgn}(\omega) = \begin{cases} -\mathrm{j}, & \omega > 0 \\ \mathrm{j}, & \omega < 0 \end{cases}$$

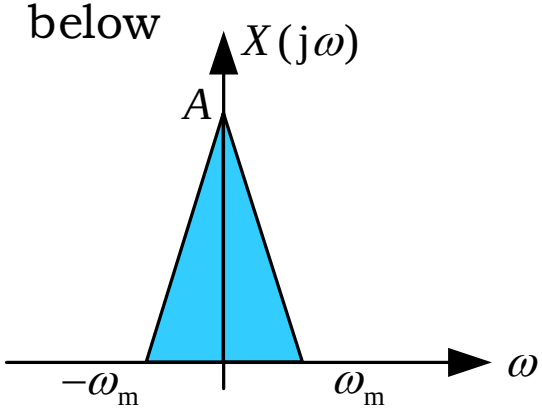
$$h(t) = \mathscr{F}^{-1}\{H(\mathrm{j}\omega)\} = \frac{1}{\pi t}$$

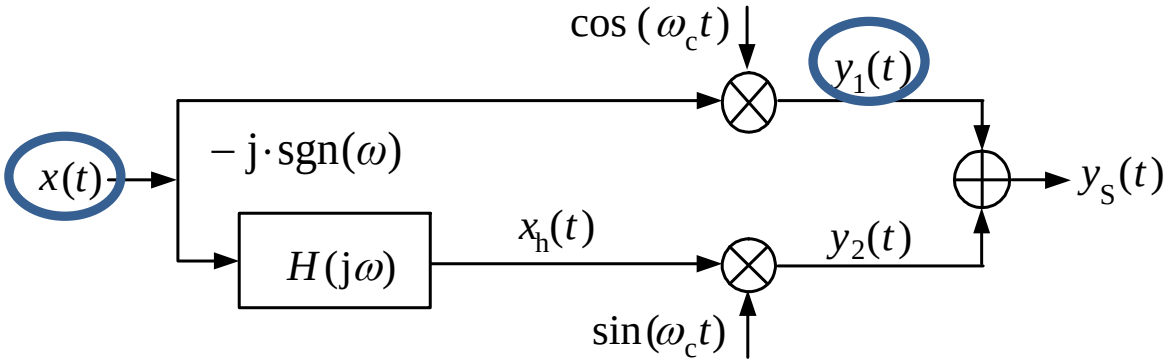




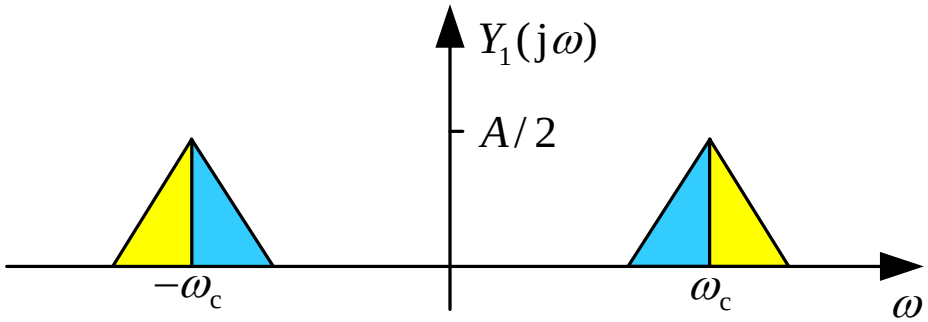
Unilateral band amplitude modulation technology

Analysis of frequency domain analysis with amplitude amplitude below



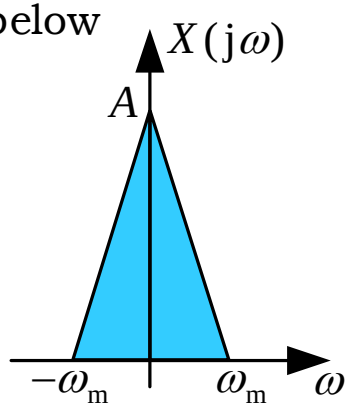


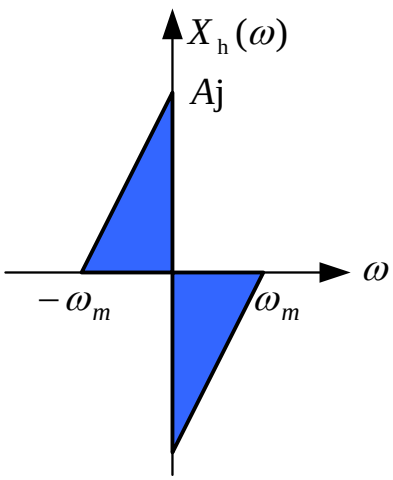
$$Y_1(\mathrm{j}\omega)=\frac{1}{2}\left\{X[\mathrm{j}(\omega+\omega_{\mathrm{c}})]+X[\mathrm{j}(\omega-\omega_{\mathrm{c}})]\right\}$$

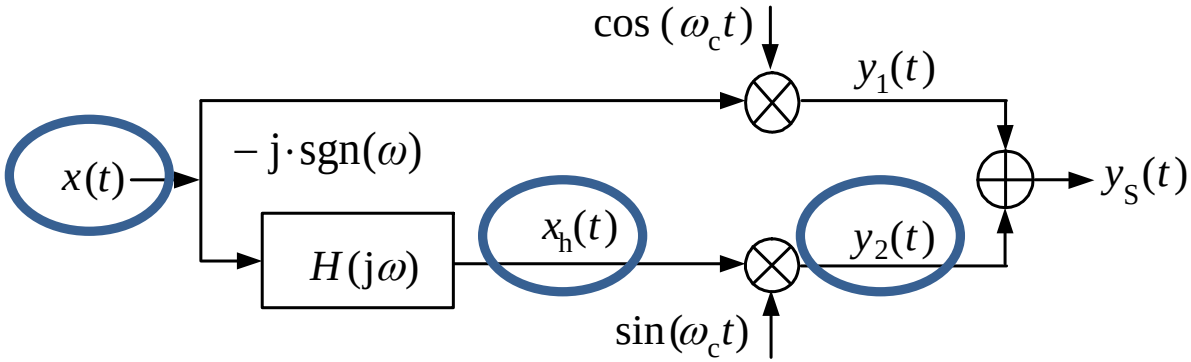


Unilateral band amplitude modulation technology

Analysis of frequency domain analysis with amplitude amplitude below

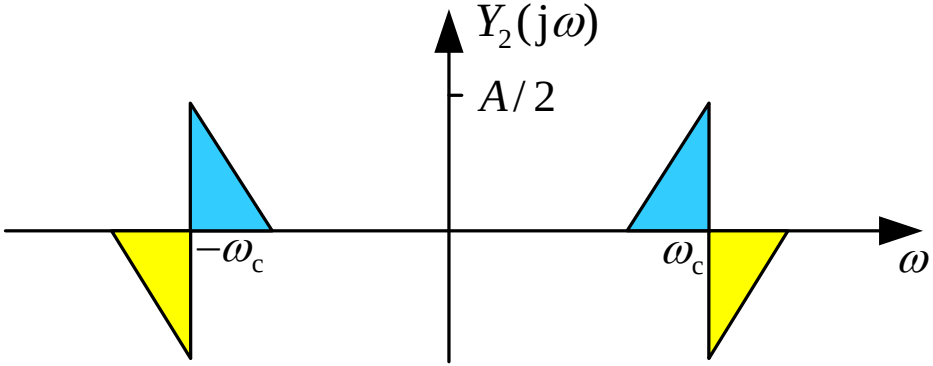






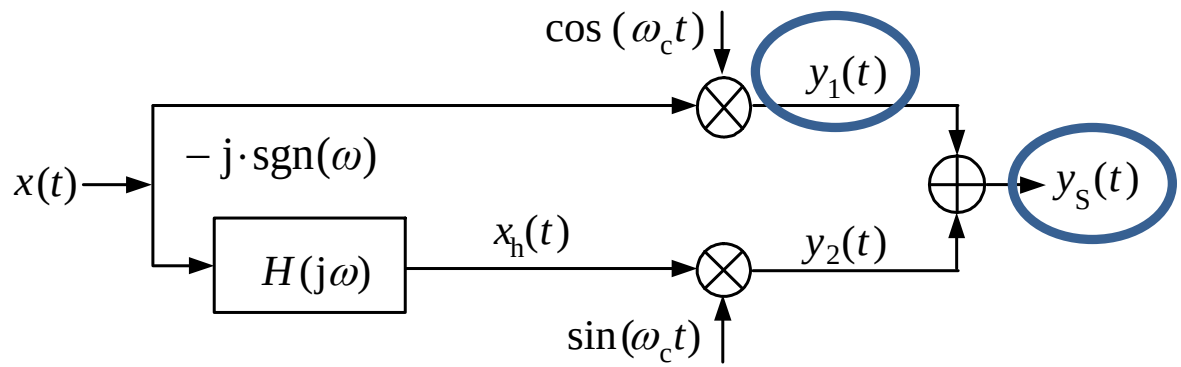
$$Y_2(\mathrm{j}\omega)=\frac{1}{2\mathrm{j}}\{X_{\mathrm{h}}[\mathrm{j}(\omega-\omega_{\mathrm{c}})]-X_{\mathrm{h}}[\mathrm{j}(\omega+\omega_{\mathrm{c}})]\}$$

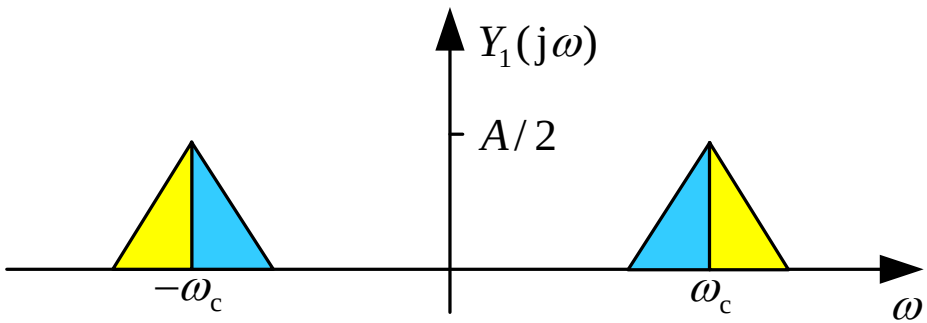
$$X_{\mathrm{h}}(\mathrm{j}\omega)=-\mathrm{j}X(\mathrm{j}\omega)\mathrm{sgn}(\omega)$$

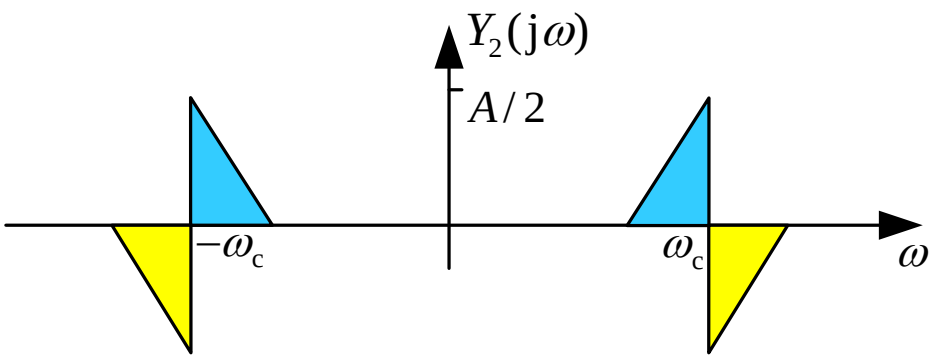


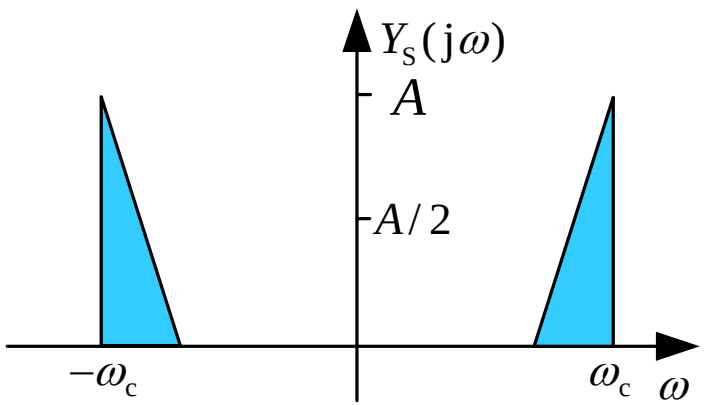
Unilateral band amplitude modulation technology

Analysis of frequency domain analysis with amplitude amplitude below







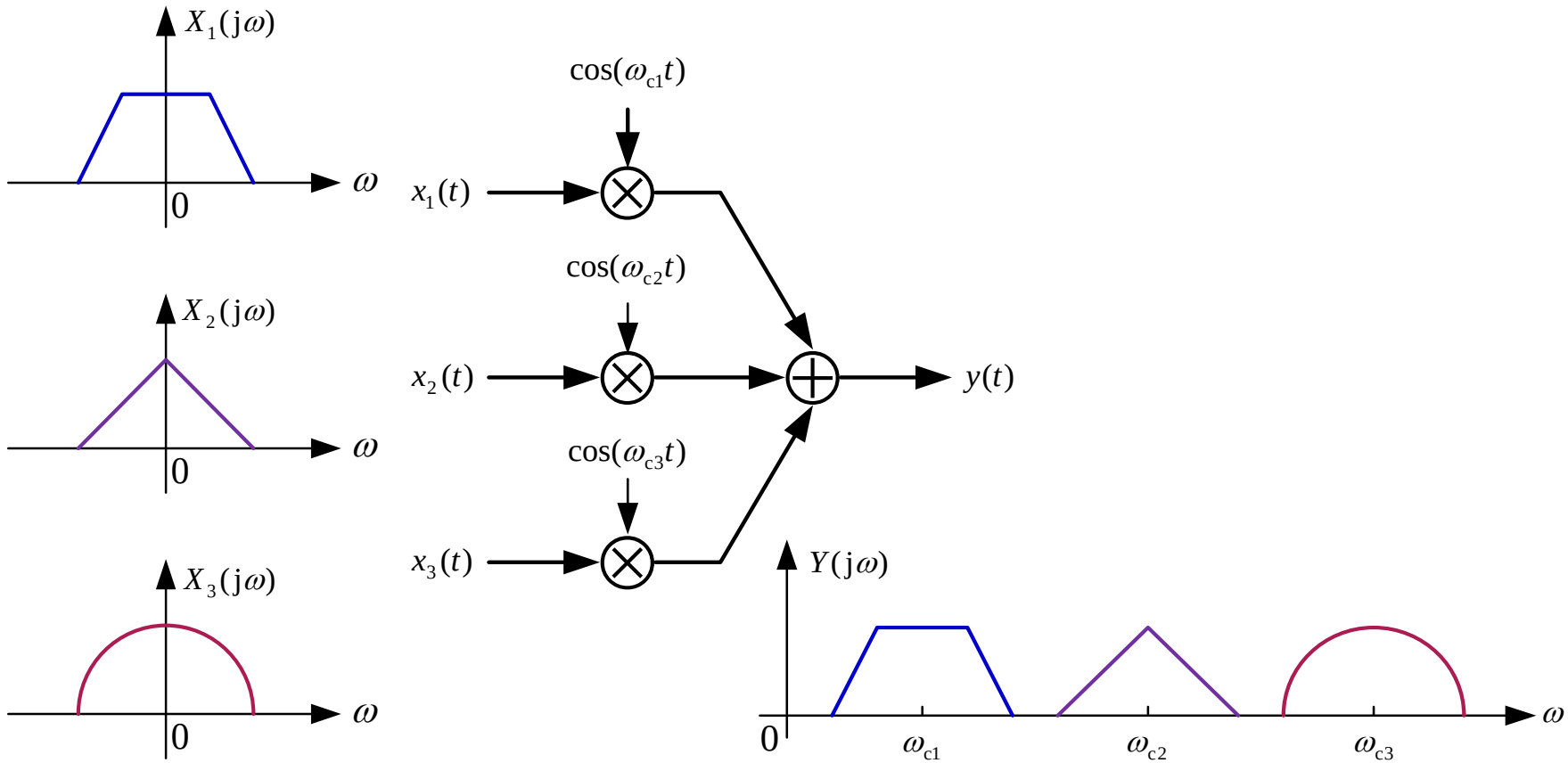


Frequent division of signal transmission

- The effective frequency band of radio broadcast signals is generally narrow, far less than the frequency bandwidth of the transmission channel
- For example, the medium wave band of the amplitude broadcast is 526.5-1606.5kHz, while the effective bandwidth of the broadcast signal is less than tens of KHz
- Frequency division reuse technology can make full use of the width of the channel

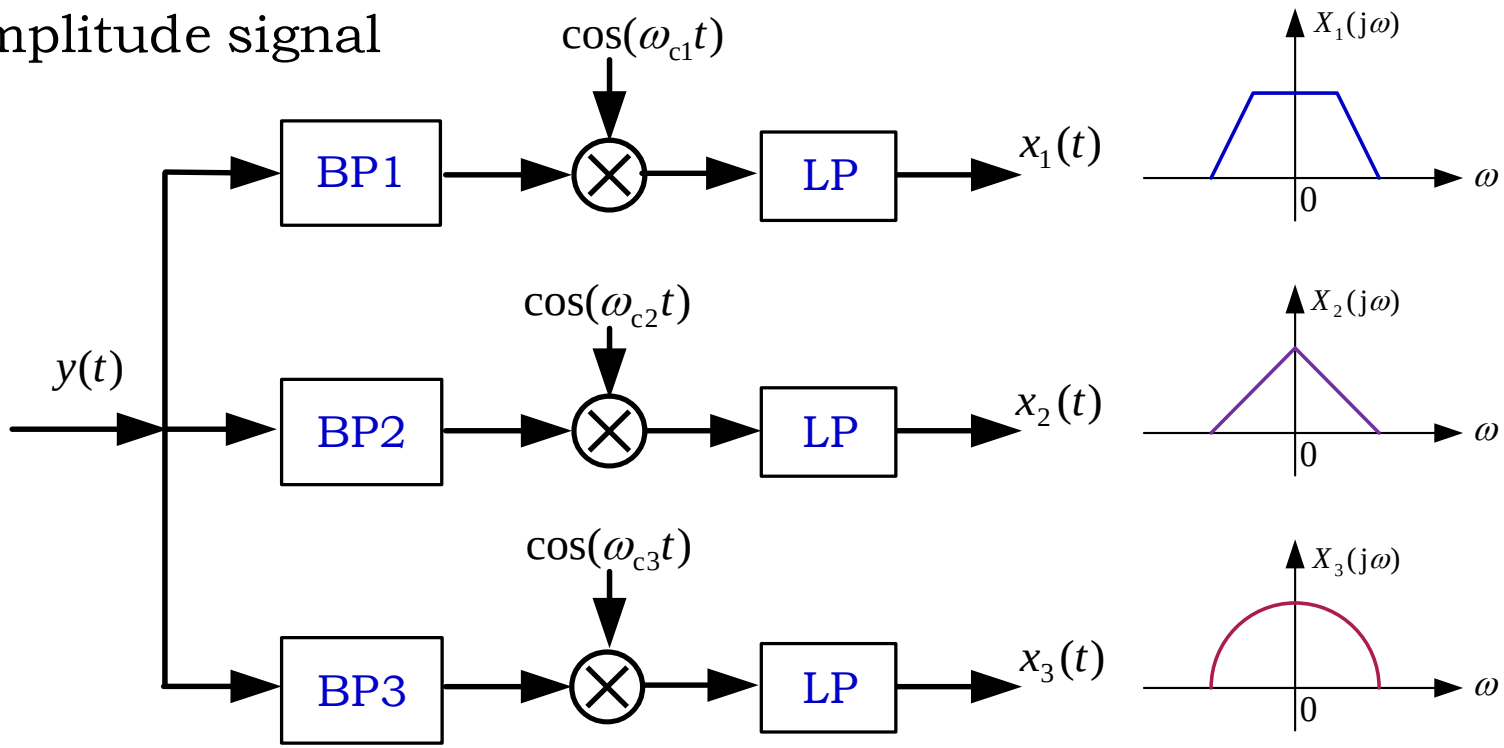
Frequent division of signal transmission

By changing the frequency of the carrier signal, different modulation signals can be moved to different frequency intervals, thereby achieving frequency duplication.



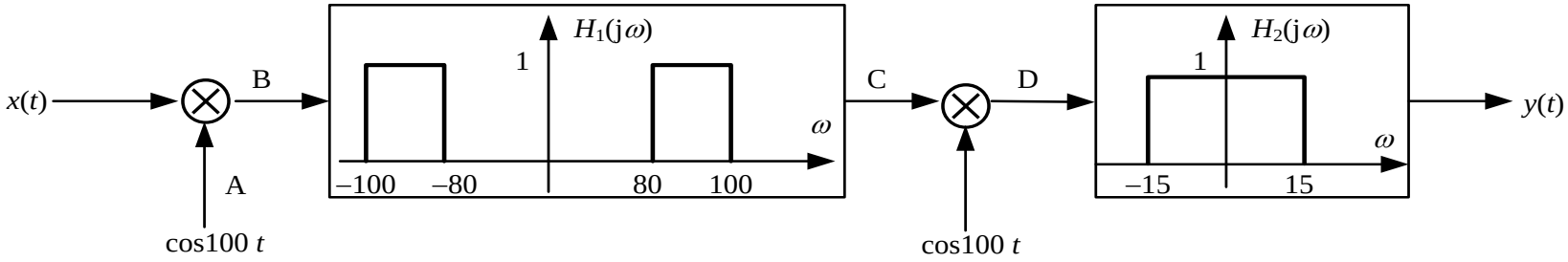
Frequent division of signal transmission

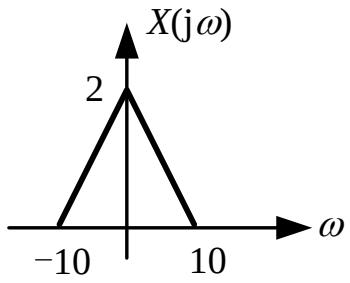
Dormation of multi -way amplitude signal

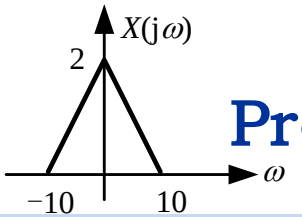


Preparation and demodulation example

[Example] As shown in the system, the spectrum $X(\mathrm{j}\omega)$ of the input signal $x(t)$ is known to analyze the spectrum of each point of A, B, C, D) in the system and drawing the spectrum chart , Seek the relationship between $y(t)$ and $x(t)$.

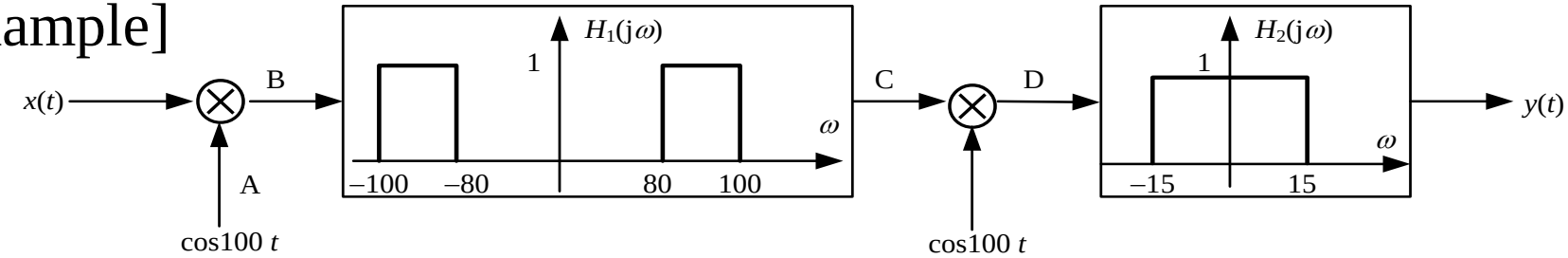






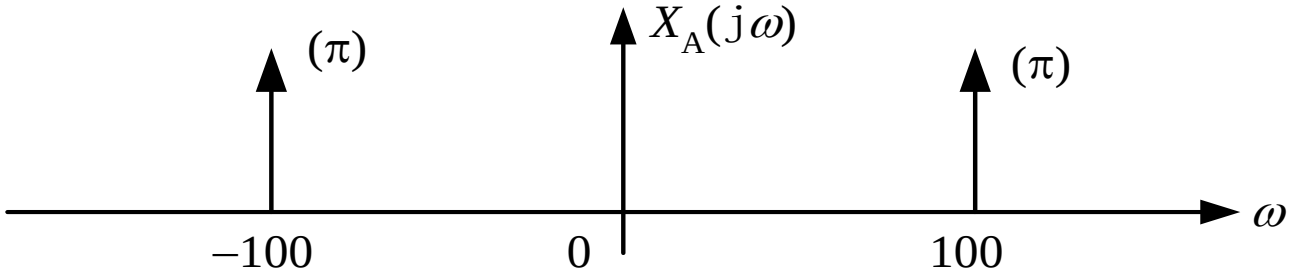
Preparation and demodulation example

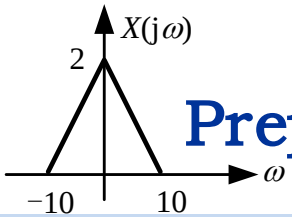
[Example]



$$\text{Solution}: X_{\mathrm{A}}(\mathrm{j}\omega) = \mathscr{F}[\cos(100t)]$$

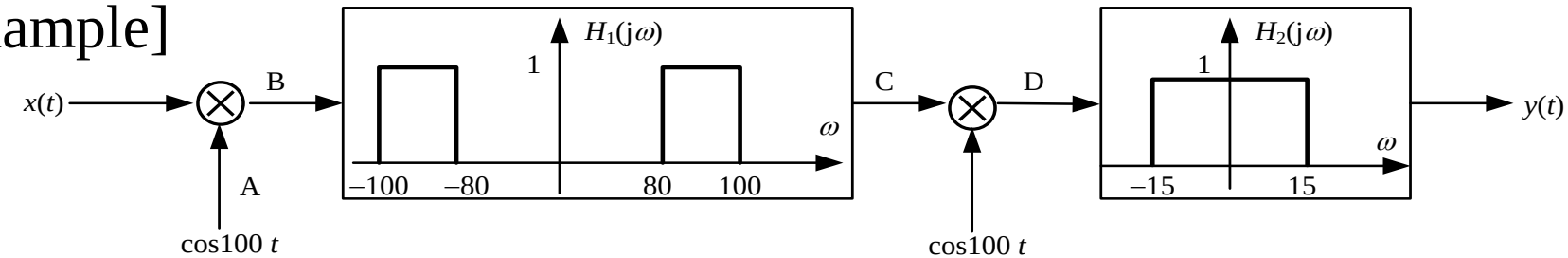
$$= \pi[\delta(\omega - 100) + \delta(\omega + 100)]$$





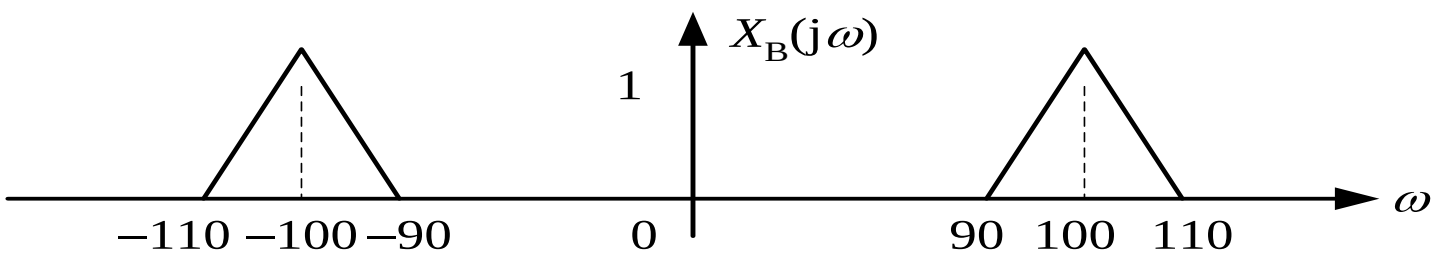
Preparation and demodulation example

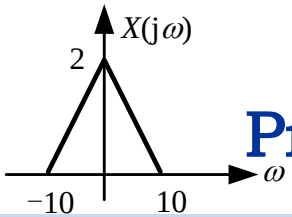
[Example]



$$\text{Solution}: X_{\mathrm{B}}(\mathrm{j}\omega) = \frac{1}{2\pi} X(\mathrm{j}\omega) * X_{\mathrm{A}}(\mathrm{j}\omega)$$

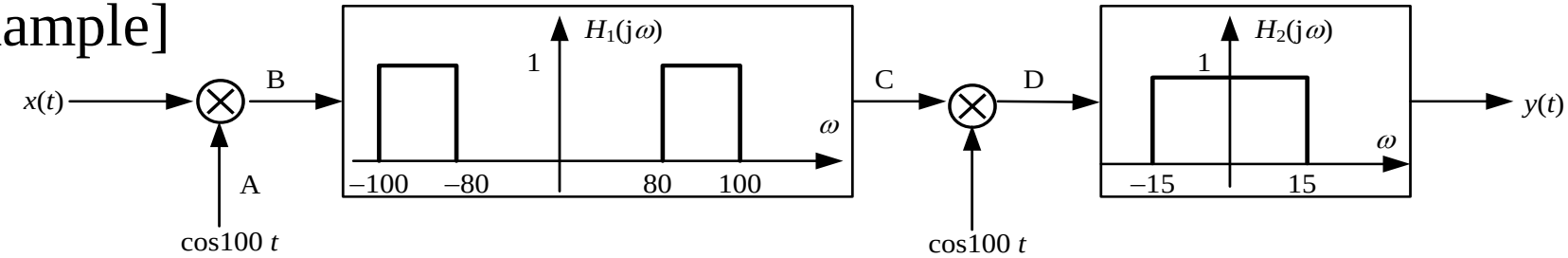
$$= \frac{1}{2}[X(\omega - 100) + X(\omega + 100)]$$





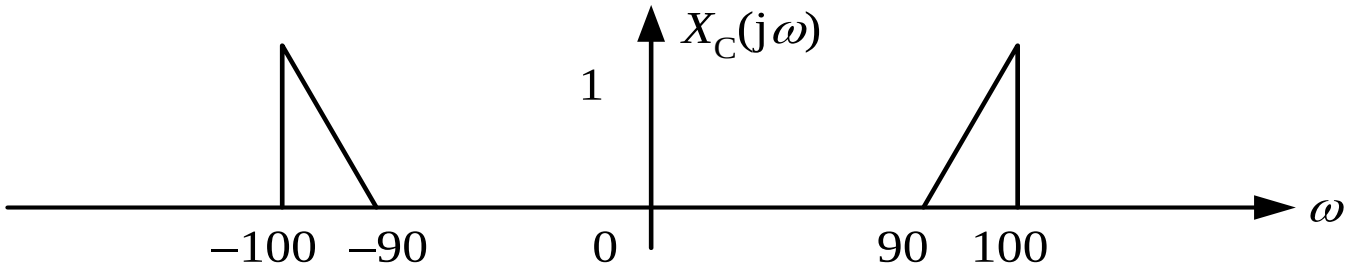
Preparation and demodulation example

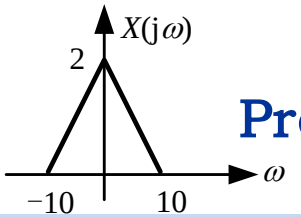
[Example]



Solution:

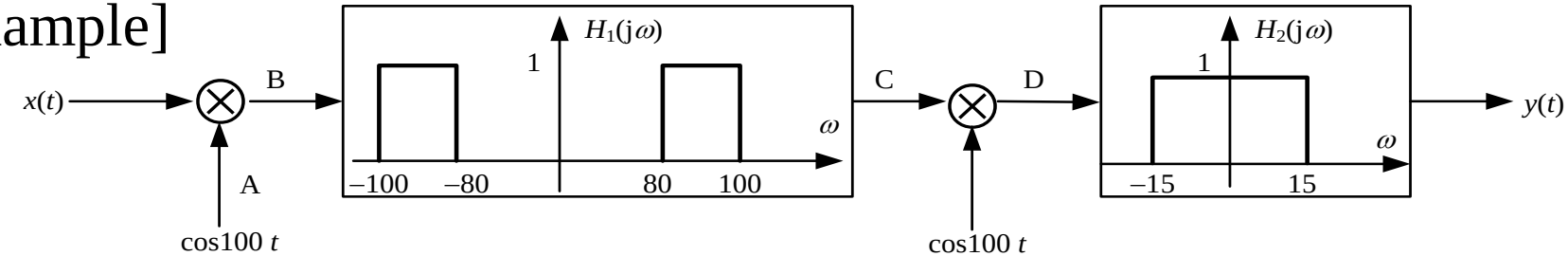
$$X_{\mathrm{C}}(\mathrm{j}\omega) = X_{\mathrm{B}}(\mathrm{j}\omega)H_1(\mathrm{j}\omega)$$



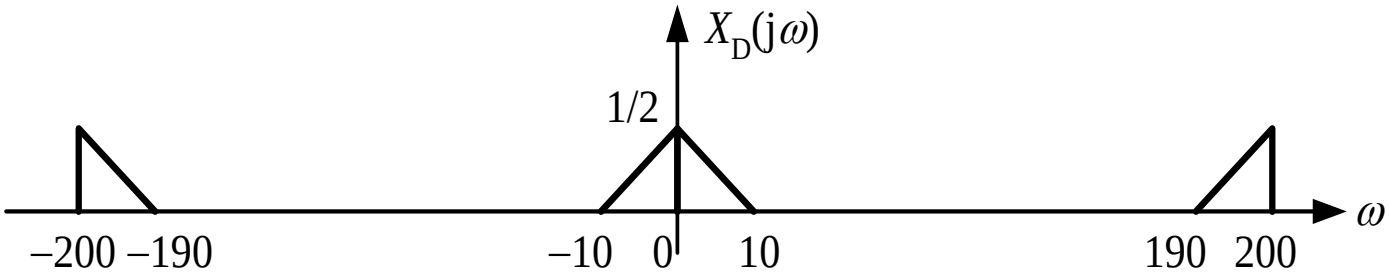


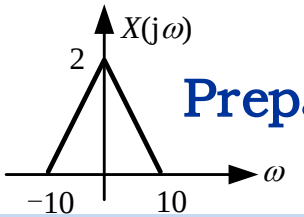
Preparation and demodulation example

[Example]



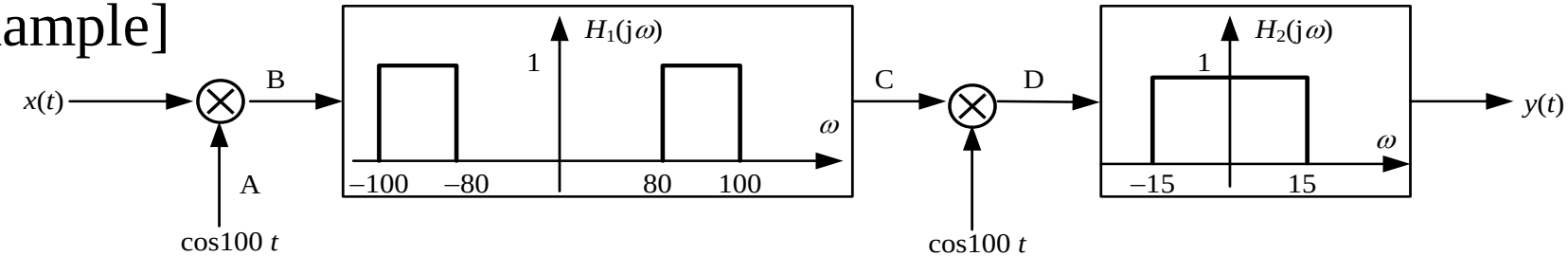
$$\text{Solution}: X_{\mathrm{D}}(\mathrm{j}\omega) = \frac{1}{2}[X_{\mathrm{C}}(\omega+100) + X_{\mathrm{C}}(\omega-100)]$$



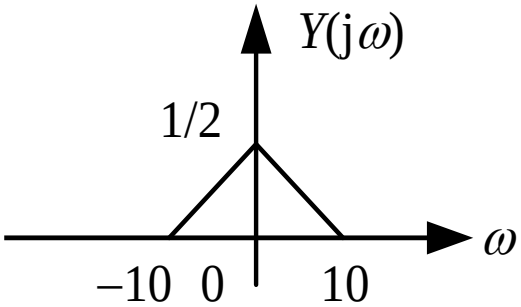


Preparation and demodulation example

[Example]



Solution: $Y(\mathrm{j}\omega) = X_{\mathrm{D}}(\mathrm{j}\omega)H_2(\mathrm{j}\omega)$



$$Y(\mathrm{j}\omega) = \frac{1}{4}X(\mathrm{j}\omega)$$

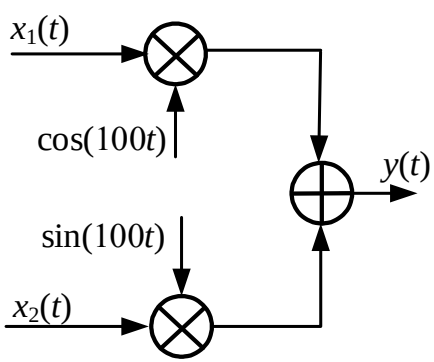
$$y(t) = \frac{1}{4}x(t)$$

Preparation and demodulation example

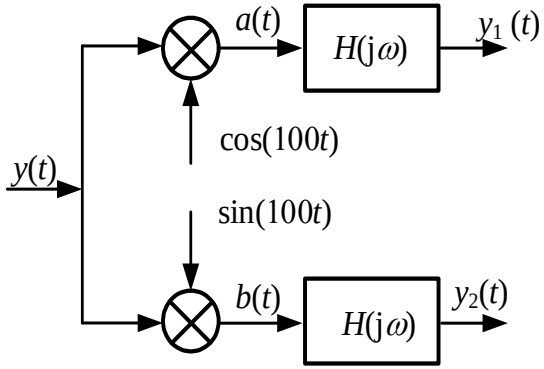
[Example] The orthogonal amplitude modulation system is shown in the figure

(1) Try the spectrum of $y(t)$、$a(t)$, and $y_1(t)$ and draw the spectrum chart;

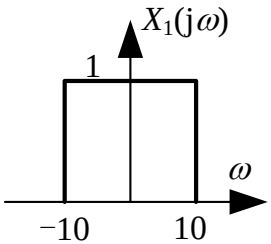
(2) Try the spectrum of $b(t)$ and $y_2(t)$ and draw the spectrum diagram.

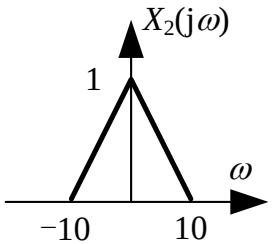


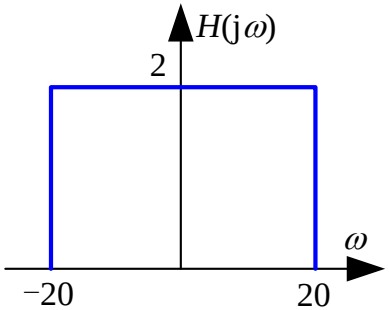
Modulator



Demodulator

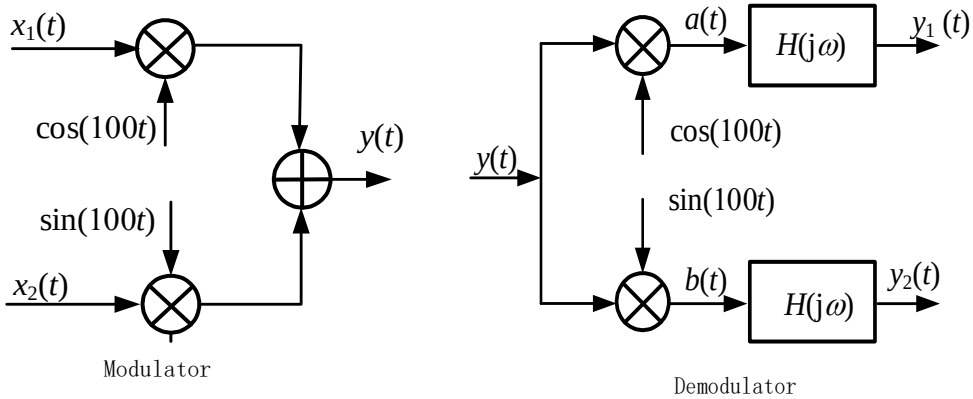






Preparation and demodulation example

[Example] orthogonal amplitude modulation system



Solution: (1) The spectrum of $y(t)$, $a(t)$, and $y_1(t)$

$$Y(\mathrm{j}\omega)=[X_1(\mathrm{j}(\omega-100))+X_1(\mathrm{j}(\omega+100))]/2$$

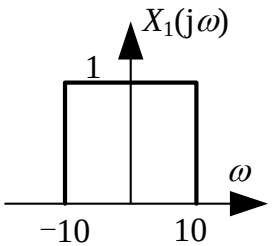
$$+[X_2(\mathrm{j}(\omega-100))-X_2(\mathrm{j}(\omega+100))]/2\mathrm{j}$$

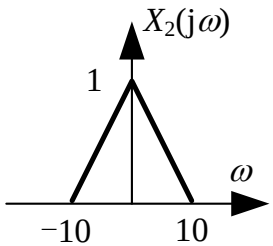
$$\mathrm{Re}\{Y(\mathrm{j}\omega)\}=\{X_1[\mathrm{j}(\omega-100)]+X_1[\mathrm{j}(\omega+100)]\}/2$$

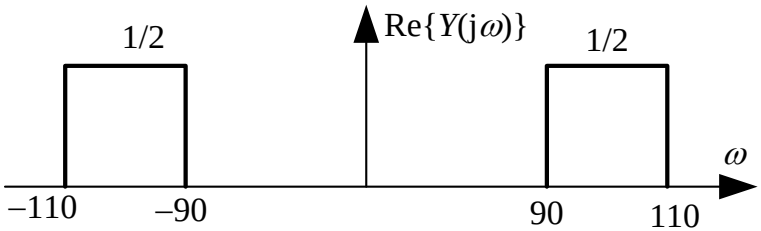
$$\mathrm{Im}\{Y(\mathrm{j}\omega)\}=\{-X_2[\mathrm{j}(\omega-100)]+X_2[\mathrm{j}(\omega+100)]\}/2$$

Preparation and demodulation example

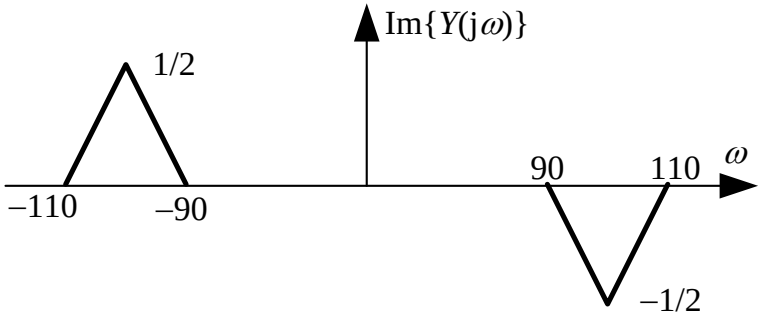
Solution: (1) The spectrum of $y(t)$、 $a(t)$, and $y_1(t)$







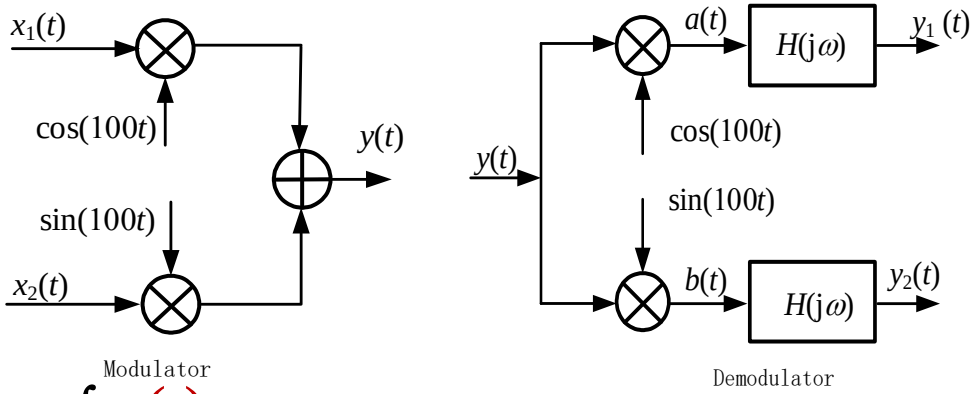
$$\mathrm{Re}\{Y(\mathrm{j}\omega)\} = \{X_1[\mathrm{j}(\omega-100)] + X_1[\mathrm{j}(\omega+100)]\}/2$$



$$\mathrm{Im}\{Y(\mathrm{j}\omega)\} = \{-X_2[\mathrm{j}(\omega-100)] + X_2[\mathrm{j}(\omega+100)]\}/2$$

Preparation and demodulation example

[Example] orthogonal amplitude modulation system



Solution: (1) The spectrum of $y(t)$、$a(t)$, and $y_1(t)$

$$A(\mathrm{j}\omega)=[Y(\mathrm{j}(\omega-100))+Y(\mathrm{j}(\omega+100))]/2$$

$$\mathrm{Re}\{A(\mathrm{j}\omega)\}=[\mathrm{Re}\{Y(\mathrm{j}(\omega-100))\}+\mathrm{Re}\{Y(\mathrm{j}(\omega+100))\}]/2$$

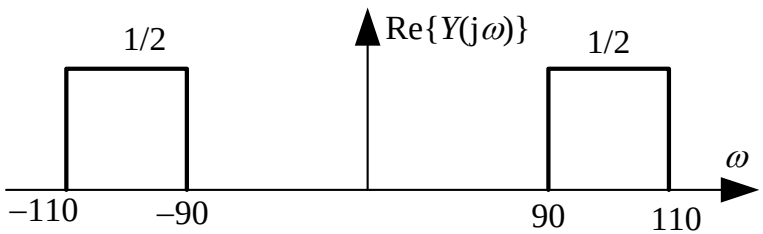
$$\mathrm{Im}\{A(\mathrm{j}\omega)\}=[\mathrm{Im}\{Y(\mathrm{j}(\omega-100))\}+\mathrm{Im}\{Y(\mathrm{j}(\omega+100))\}]/2$$

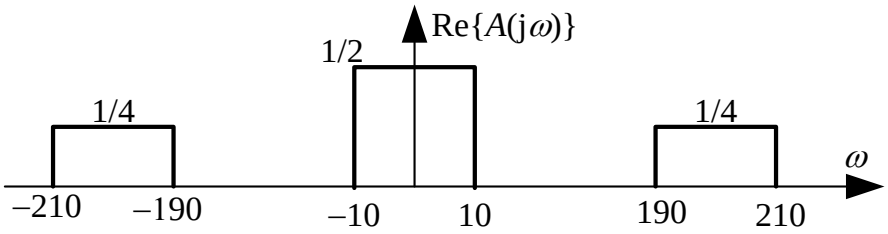
Preparation and demodulation example

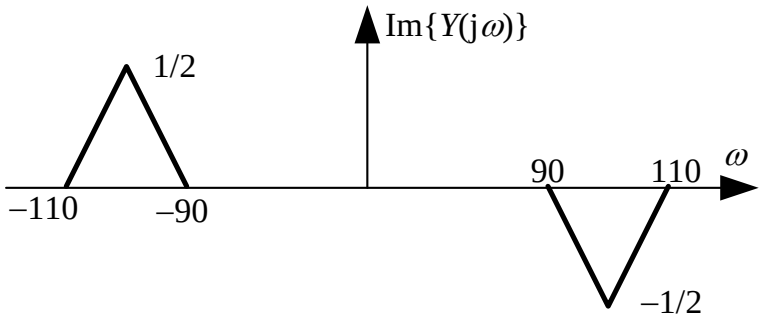
Solution: (1) The spectrum of $y(t)$、 $a(t)$, and $y_1(t)$

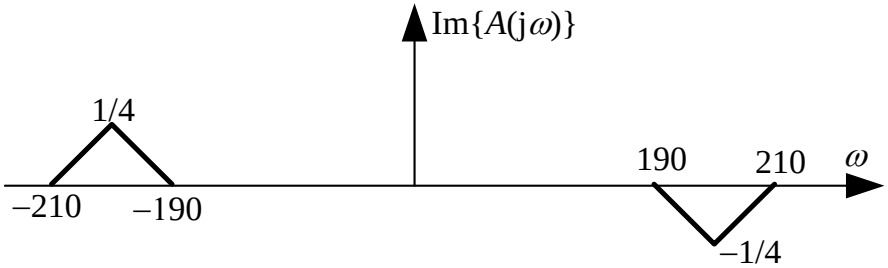
$$\mathrm{Re}\{A(\mathrm{j}\omega)\} = [\mathrm{Re}\{Y(\mathrm{j}(\omega-100))\} + \mathrm{Re}\{Y(\mathrm{j}(\omega+100))\}]/2$$

$$\mathrm{Im}\{A(\mathrm{j}\omega)\} = [\mathrm{Im}\{Y(\mathrm{j}(\omega-100))\} + \mathrm{Im}\{Y(\mathrm{j}(\omega+100))\}]/2$$



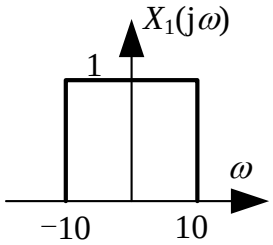


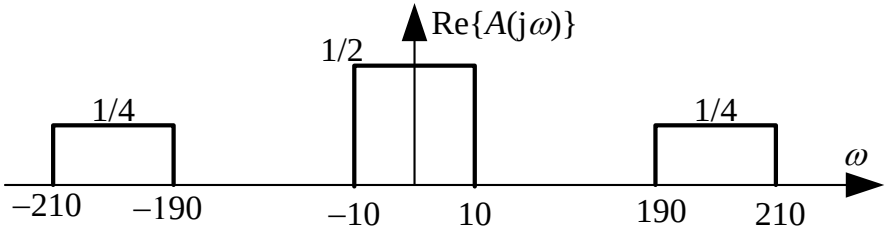


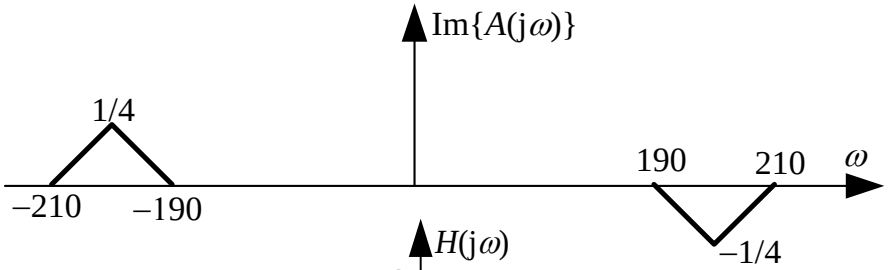


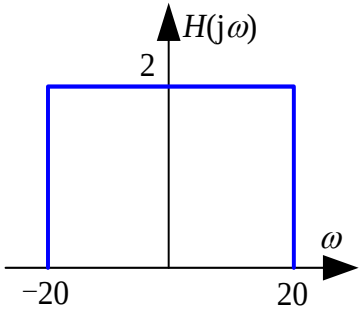
Preparation and demodulation example

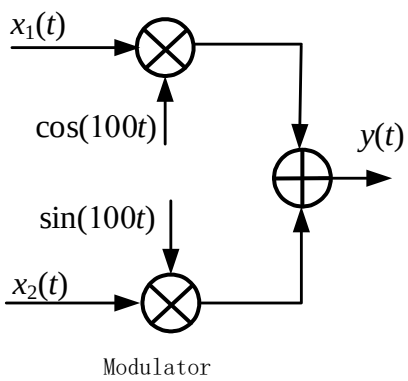
Solution: (1) The spectrum of $y(t)$、$a(t)$, and $y_1(t)$

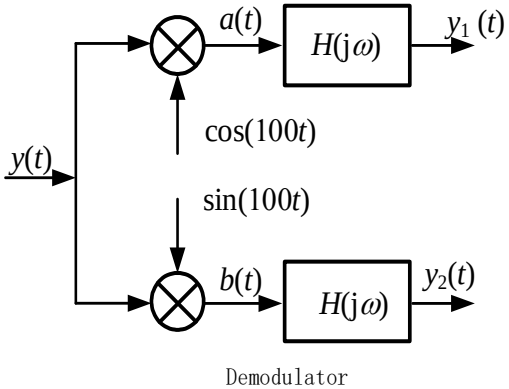




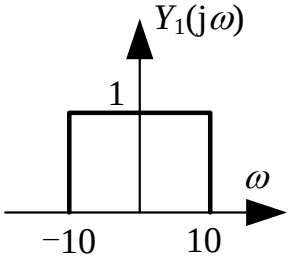








$$Y_1(\mathrm{j}\omega) = H(\mathrm{j}\omega)A(\mathrm{j}\omega)$$

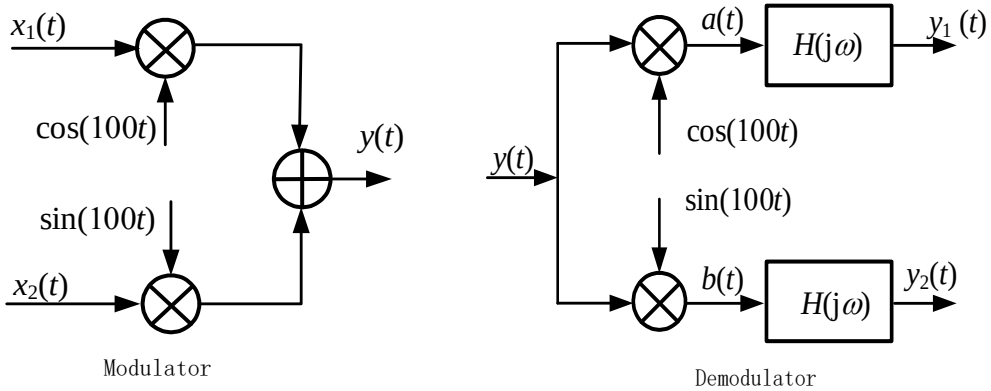


So $Y_1(\mathrm{j}\omega) = X_1(\mathrm{j}\omega)$

Preparation and demodulation example

[Example] orthogonal amplitude modulation system

Solution: (2) spectrum of (2) $b(t)$ and $y_2(t)$



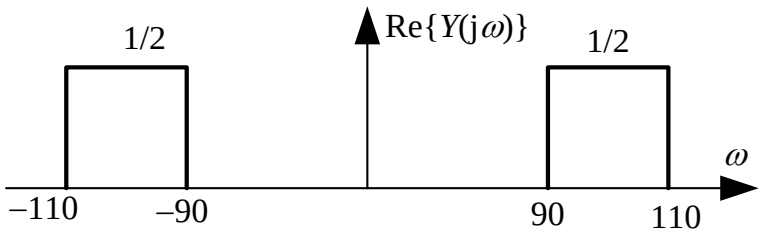
$$B(\mathrm{j}\omega)=[Y(\mathrm{j}(\omega-100))-Y(\mathrm{j}(\omega+100))]/2\mathrm{j}$$

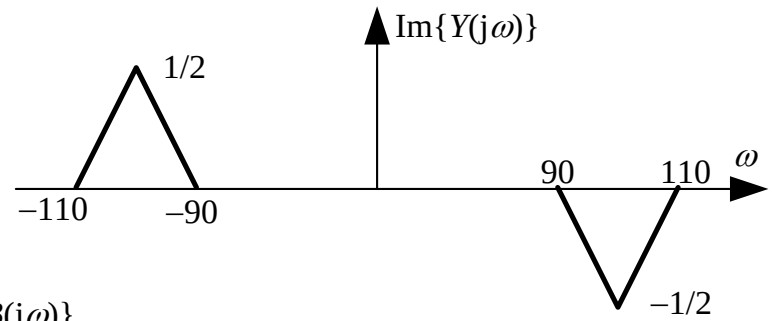
$$\mathrm{Re}\{B(\mathrm{j}\omega)\}=[\mathrm{Im}\{Y(\mathrm{j}(\omega-100))\}-\mathrm{Im}\{Y(\mathrm{j}(\omega+100))\}]/2$$

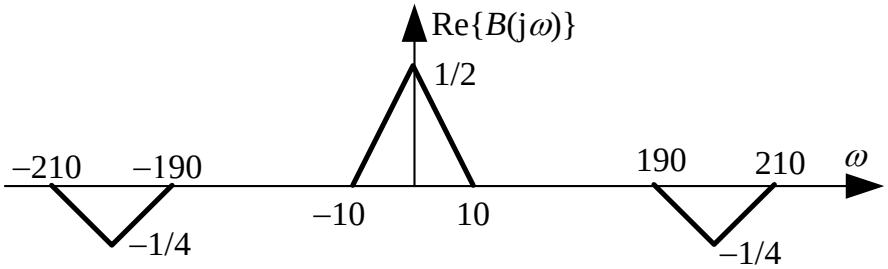
$$\mathrm{Im}\{B(\mathrm{j}\omega)\}=[-\mathrm{Re}\{Y(\mathrm{j}(\omega-100))\}+\mathrm{Re}\{Y(\mathrm{j}(\omega+100))\}]/2$$

Preparation and demodulation example

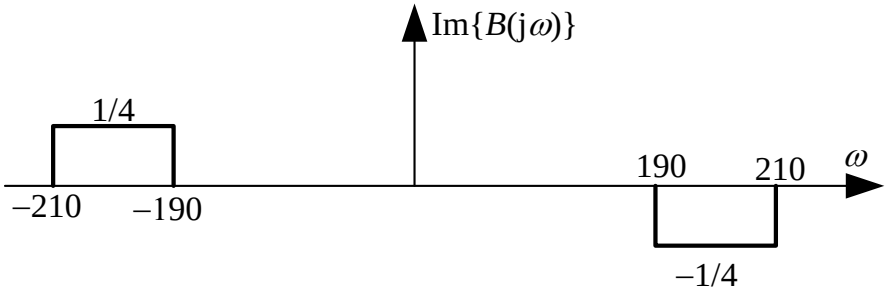
Solution: (2) spectrum of $b(t)$ and $y_2(t)$







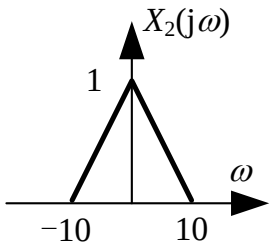
$$\mathrm{Re}\{B(\mathrm{j}\omega)\} = [\mathrm{Im}\{Y(\mathrm{j}(\omega-100))\} - \mathrm{Im}\{Y(\mathrm{j}(\omega+100))\}]/2$$

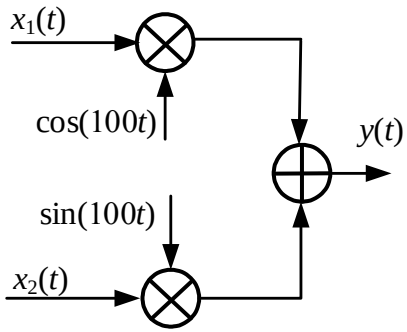


$$\mathrm{Im}\{B(\mathrm{j}\omega)\} = [-\mathrm{Re}\{Y(\mathrm{j}(\omega-100))\} + \mathrm{Re}\{Y(\mathrm{j}(\omega+100))\}]/2$$

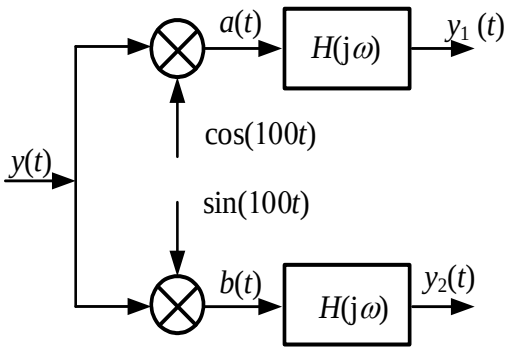
Preparation and demodulation example

Solution: (2) spectrum of $b(t)$ and $y_2(t)$

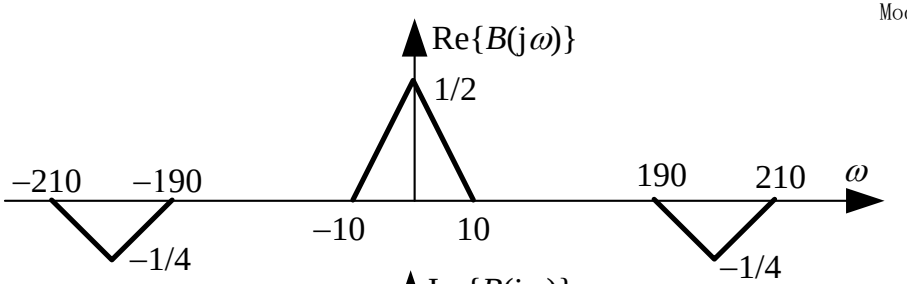


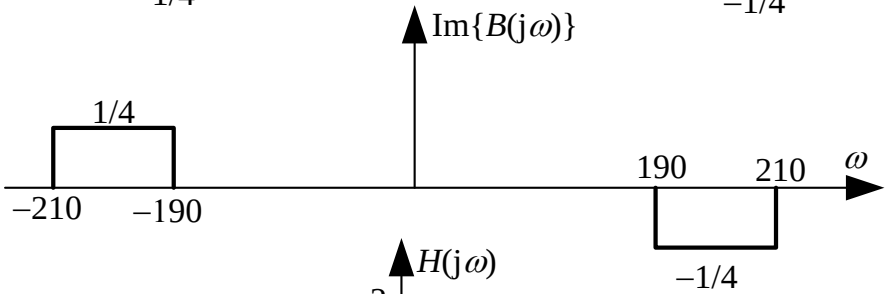


Modulator

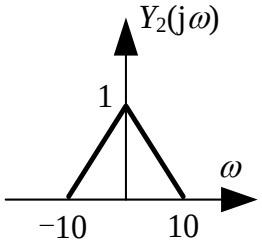


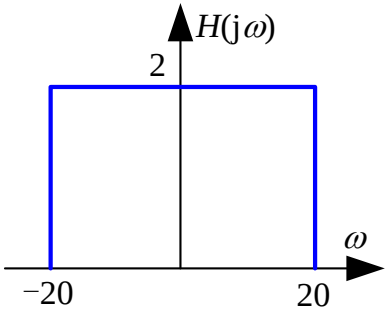
Demodulator





$$Y_2(\mathrm{j}\omega)\} = H(\mathrm{j}\omega)B(\mathrm{j}\omega)$$





So $\quad Y_2(\mathrm{j}\omega) = X_2(\mathrm{j}\omega)$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!